

BOOK DATA ANALYSIS USING HADOOP



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Problem Statement: Analyze the BookDataset using Hadoop(MapReduce), Pig based on different column fields.

Dataset Description:

Dataset: <https://www.kaggle.com/datasets/arashnic/book-recommendation-dataset>

Books.csv:

- **ISBN:** Unique identifier string for each book.
- **Book-Title / Book-Author:** String fields for title and primary author.
- **Year-Of-Publication:** Integer representing publication year.
- **Publisher:** Publishing house name.
- **Image-URL-S / M / L:** Cover image links (Small, Medium, Large).

Users.csv:

- **User-ID:** Anonymized integer user identifier.
- **Location:** User location string.
- **Age:** Integer age of user (may contain nulls).

Ratings.csv:

- **User-ID / ISBN:** Composite key mapping user to book.
- **Book-Rating:** Rating score (Explicit: 1–10, Implicit: 0).

Summary of Data Analysis:

Following MapReduce programs and its different design patterns like mentioned below are implemented:

1. Filtering
2. Join Patterns
3. Data Organization
4. Summarization

Following analysis can be performed on the BookDataset:

1. **Rating Summarization:** Calculate Minimum Rating, Maximum Rating, Total Rating, Rating Count and Average Rating for each ISBN.
2. **Top Books:** Identify the top books based on the number of ratings/reviews.
3. **Top Books by Publisher:** Identify the top books for each publisher based on the number of ratings/reviews.

Following Pig analysis is performed on the BookDataset:

1. Calculate the Top 5 Publishers based on the number of distinct books.
2. Calculate the Top 10 Most Reviewed Books.
3. Calculate the Top 10 Most Reviewed Books by Publisher.
4. Calculate the Top 10 Highest Rated Books.
5. Calculate the Top 10 Highest Rated Books by Publisher.

Map Reduce Analysis

Merge Files: Merges Books.csv and Ratings.csv based on the common ISBN field and generates a single BookDataset.csv file, which is stored in HDFS.

BookDataset

- **ISBN:** Unique identifier string for each book.
- **BookTitle:** String field containing the title of the book.
- **Author:** String field containing the primary author of the book.
- **Year:** Integer representing the publication year.
- **Publisher:** String field containing the publisher name.
- **UserID:** Unique identifier for each user who rated the book.
- **Rating:** Integer representing the rating given by the user.

Code

```
1 package mergedataset;
2
3 import java.io.*;
4 import java.util.*;
5
6 /** Merges Goodreads Books.csv with Ratings.csv using ISBN and writes the classroom-friendly bookDataset.csv. */
7 public class MergeCSV {
8     static String[] parse(String line){List<String> f=new ArrayList<String>();StringBuilder s=new StringBuilder();boolean q=false;for(int i=0;i<line.length();i++){char c
9     static String clean(String s){return s.replace('\t',' ').replace('\r',' ').replace('\n',' ').replace(',',' ').trim();}
10    public static void main(String[] args)throws Exception{
11        if(args.length!=3){System.err.println("Usage: MergeCSV <Books.csv> <Ratings.csv> <bookDataset.csv>");System.exit(2);}
12        Map<String,String[]> books=new HashMap<String,String[]>();
13        BufferedReader br=new BufferedReader(new InputStreamReader(new FileInputStream(args[0]),"UTF-8"));
14        String line; boolean first=true; while((line=br.readLine())!=null){if(first){first=false;continue;}String[] f=parse(line);if(f.length==8){String isbn=f[0].trim()
15        BufferedReader rr=new BufferedReader(new InputStreamReader(new FileInputStream(args[1]),"UTF-8"));
16        BufferedWriter out=new BufferedWriter(new OutputStreamWriter(new FileOutputStream(args[2]),"UTF-8"));
17        out.write("ISBN,BookTitle,Author,Year,Publisher,UserID,Rating\n"); first=true; long rows=0;
18        while((line=rr.readLine())!=null){if(first){first=false;continue;}String[] r=parse(line);if(r.length<3)continue;String[] b=books.get(r[1].trim());if(b==null)cont
19            try{Float.parseFloat(r[2].trim());}catch(Exception e){continue;}
20            out.write(clean(b[0])+" "+clean(b[1])+" "+clean(b[2])+" "+clean(b[3])+" "+clean(b[4])+" "+clean(r[0])+" "+clean(r[2])+" \n");rows++;}
21        rr.close();out.close();System.out.println("Merged rows: "+rows);
22    }
23 }
24
```

Execution

```
C:\BDLFinalProject\mergedataset\src\main\java>java -cp . mergedataset.MergeCSV C:\BDLFinalProject\rawdata\Books.csv C:\BDLFinalProject\rawdata\Ratings.csv C:\BDLFinalProject\data\bookDataset.csv
Merged rows: 1031139

C:\BDLFinalProject\mergedataset\src\main\java>dir C:\BDLFinalProject\data
Volume in drive C has no label.
Volume Serial Number is 3C45-1A14

Directory of C:\BDLFinalProject\data

09/12/2026  07:19 PM    <DIR>          .
09/12/2026  09:48 AM    <DIR>          ..
09/12/2026  07:19 PM             91,219,732 bookDataset.csv
09/10/2026  10:14 AM             73,293,635 Books.csv
09/10/2026  10:14 AM             22,633,892 Ratings.csv
09/12/2026  09:48 AM               143 README.txt
09/10/2026  10:13 AM             11,017,438 Users.csv
               5 File(s)          198,164,840 bytes
               2 Dir(s)        13,139,173,376 bytes free
```

1. Rating Summarization: Calculate Minimum Rating, Maximum Rating, Total Rating, Rating Count and Average Rating for each ISBN.

This analysis comprises of:

- **Driver class:** Sets up the configuration for the Mapper, Reducer and Tuple classes.
- **Mapper class:** Reads each rating record and emits ISBN as the key and the rating information as the value.
- **Reducer class:** Groups ratings by ISBN and calculates the Minimum Rating, Maximum Rating, Total Rating, Rating Count and Average Rating for each book.
- **Tuple class:** A custom Writable class used to store and transfer the minimum rating, maximum rating, total rating, rating count and average rating values between the Mapper and Reducer.

Mapper

```
J MapperClass.java  J Rating_Summarization.java  J RatingSummaryTuple.java  J ReducerClass.java  J CsvUtils.java
C: > BDLFinalProject > Rating_Summarization > src > main > java > com > neu > bigdata > J MapperClass.java
1  package com.neu.bigdata;
2
3  import java.io.IOException;
4  import org.apache.hadoop.io.Text;
5  import org.apache.hadoop.mapreduce.Mapper;
6
7  public class MapperClass extends Mapper<Object, Text, Text, RatingSummaryTuple> {
8      private final Text isbn = new Text();
9      private final RatingSummaryTuple tuple = new RatingSummaryTuple();
10     protected void map(Object key, Text value, Context context) throws IOException, InterruptedException {
11         String line=value.toString();
12         if(line.trim().isEmpty()) return;
13         String[] f=CsvUtils.parseLine(line);
14         if(CsvUtils.isHeader(f) || f.length < 7) return;
15         try {
16             float rating=Float.parseFloat(f[6].trim());
17             if(f[0].trim().isEmpty()) return;
18             isbn.set(f[0].trim()); tuple.add(rating); context.write(isbn,tuple);
19             // Writable instances are reused by Hadoop, so reset for the next record.
20             tuple.setFrom(new RatingSummaryTuple());
21         } catch(NumberFormatException ignored) {}
22     }
23 }
```

Reducer

```
J Rating_Summarization.java  J RatingSummaryTuple.java  J ReducerClass.java  J CsvUtils.java
C: > BDLFinalProject > Rating_Summarization > src > main > java > com > neu > bigdata > J ReducerClass.java
1  package com.neu.bigdata;
2
3  import java.io.IOException;
4  import org.apache.hadoop.io.Text;
5  import org.apache.hadoop.mapreduce.Reducer;
6
7  public class ReducerClass extends Reducer<Text, RatingSummaryTuple, Text, RatingSummaryTuple> {
8      protected void reduce(Text key, Iterable<RatingSummaryTuple> values, Context context) throws IOException, InterruptedException {
9          RatingSummaryTuple result=new RatingSummaryTuple();
10         for(RatingSummaryTuple v:values) result.merge(v);
11         context.write(key,result);
12     }
13 }
```

Tuple

```
J Rating_Summarization.java J RatingSummaryTuple.java J CsvUtils.java
C:\> BDLFinalProject > Rating_Summarization > src > main > java > com > neu > bigdata > J RatingSummaryTuple.java

8 public class RatingSummaryTuple implements Writable {
9     private float minRating = 0f;
10    private long ratingCount = 0L;
11
12    public RatingSummaryTuple() {}
13    public RatingSummaryTuple(float rating) { add(rating); }
14    public void add(float rating) {
15        if (rating < minRating) minRating = rating;
16        if (rating > maxRating) maxRating = rating;
17        totalRating += rating;
18        ratingCount++;
19    }
20    public void merge(RatingSummaryTuple other) {
21        if (other.ratingCount == 0) return;
22        if (ratingCount == 0 || other.minRating < minRating) minRating = other.minRating;
23        if (ratingCount == 0 || other.maxRating > maxRating) maxRating = other.maxRating;
24        totalRating += other.totalRating;
25        ratingCount += other.ratingCount;
26    }
27    public void setFrom(RatingSummaryTuple other){ minRating=other.minRating; maxRating=other.maxRating; totalRating=other.totalRating; ratingCount=other.ratingCount; }
28    public float getMinRating(){return minRating;}
29    public float getMaxRating(){return maxRating;}
30    public float getTotalRating(){return totalRating;}
31    public long getRatingCount(){return ratingCount;}
32    public void write(DataOutput out) throws IOException {
33        out.writeFloat(minRating); out.writeFloat(maxRating); out.writeFloat(totalRating); out.writeLong(ratingCount);
34    }
35    public void readFields(DataInput in) throws IOException {
36        minRating=in.readFloat(); maxRating=in.readFloat(); totalRating=in.readFloat(); ratingCount=in.readLong();
37    }
38    public String toString(){
39        float avg = ratingCount == 0 ? 0f : totalRating / ratingCount;
40        return minRating + "\t" + maxRating + "\t" + totalRating + "\t" + ratingCount + "\t" + avg;
41    }
42 }
```

Driver

```
C:\> BDLFinalProject > Rating_Summarization > src > main > java > com > neu > bigdata > J Rating_Summarization.java

1 package com.neu.bigdata;
2
3 import org.apache.hadoop.conf.Configuration;
4 import org.apache.hadoop.fs.Path;
5 import org.apache.hadoop.io.Text;
6 import org.apache.hadoop.mapreduce.Job;
7 import org.apache.hadoop.mapreduce.lib.input.FileInputFormat;
8 import org.apache.hadoop.mapreduce.lib.output.FileOutputFormat;
9
10 public class Rating_Summarization {
11     public static void main(String[] args) throws Exception {
12         if(args.length!=2){System.err.println("Usage: Rating_Summarization <input> <output>");System.exit(2);}
13         Job job=Job.getInstance(new Configuration(),"Book Rating Summarization");
14         job.setJarByClass(Rating_Summarization.class);
15         job.setMapperClass(MapperClass.class); job.setCombinerClass(ReducerClass.class); job.setReducerClass(ReducerClass.class);
16         job.setOutputKeyClass(Text.class); job.setOutputValueClass(RatingSummaryTuple.class);
17         FileInputFormat.addInputPath(job,new Path(args[0])); FileOutputFormat.setOutputPath(job,new Path(args[1]));
18         System.exit(job.waitForCompletion(true)?0:1);
19     }
20 }
```

Execution

```
C:\BDLFinalProject>mkdir build1
C:\BDLFinalProject>cd C:\BDLFinalProject\Rating_Summarization\src\main\java
C:\BDLFinalProject\Rating_Summarization\src\main\java>javac -cp "C:\hadoop\share\hadoop\common\*;C:\hadoop\share\hadoop\common\lib\*;C:\hadoop\share\hadoop\hdfs\*;C:\hadoop\share\hadoop\mapreduce\*;C:\hadoop\share\hadoop\mapreduce\lib\*" -d C:\BDLFinalProject\build1 com\neu\bigdata\*.java
C:\BDLFinalProject\Rating_Summarization\src\main\java>dir C:\BDLFinalProject\build1
Volume in drive C has no label.
Volume Serial Number is 3C45-1A14

Directory of C:\BDLFinalProject\build1

09/12/2026  08:17 PM    <DIR>          .
09/12/2026  08:16 PM    <DIR>          ..
09/12/2026  08:17 PM    <DIR>          com
               0 File(s)              0 bytes
               3 Dir(s)  12,690,722,816 bytes free

C:\BDLFinalProject\build1>hadoop jar C:\BDLFinalProject\build1\ratingsummary.jar com.neu.bigdata.Rating_Summarization /BDLFinalProject/data/bookDataset.csv /BDLFinalProject/data/outputFiles/ratingSummary
1026-09-12 21:40:17,883 INFO client.RMPProxy: Connecting to ResourceManager at /0.0.0.0:8032
Exception in thread "main" org.apache.hadoop.mapred.FileAlreadyExistsException: Output directory hdfs://localhost:9000/BDLFinalProject/data/outputFiles/ratingSummary already exists
    at org.apache.hadoop.mapreduce.lib.output.FileOutputFormat.checkOutputSpecs(FileOutputFormat.java:164)
    at org.apache.hadoop.mapreduce.JobSubmitter.checkSpecs(JobSubmitter.java:277)
    at org.apache.hadoop.mapreduce.JobSubmitter.submitJobInternal(JobSubmitter.java:143)
    at org.apache.hadoop.mapreduce.Job$11.run(Job.java:1565)
    at org.apache.hadoop.mapreduce.Job$11.run(Job.java:1562)
    at java.security.AccessController.doPrivileged(Native Method)
    at javax.security.auth.Subject.doAs(Subject.java:422)
    at org.apache.hadoop.security.UserGroupInformation.doAs(UserGroupInformation.java:1762)
    at org.apache.hadoop.mapreduce.Job.submit(Job.java:1562)
    at org.apache.hadoop.mapreduce.Job.waitForCompletion(Job.java:1583)
    at com.neu.bigdata.Rating_Summarization.main(Rating_Summarization.java:18)
    at sun.reflect.NativeMethodAccessorImpl.invoke0(Native Method)
    at sun.reflect.NativeMethodAccessorImpl.invoke(NativeMethodAccessorImpl.java:62)
    at sun.reflect.DelegatingMethodAccessorImpl.invoke(DelegatingMethodAccessorImpl.java:43)
    at java.lang.reflect.Method.invoke(Method.java:498)
    at org.apache.hadoop.util.RunJar.run(RunJar.java:323)
    at org.apache.hadoop.util.RunJar.main(RunJar.java:236)
```

Output

```
C:\BDLFinalProject\build1>hadoop fs -cat /BDLFinalProject/data/outputFiles/ratingSummary/part-r-00000 | more
0000913154      8.0      8.0      8.0      1      8.0
0001010565      0.0      1.4E-45  0.0      2      0.0
0001046438      9.0      9.0      9.0      1      9.0
0001046713      0.0      1.4E-45  0.0      1      0.0
000104687X      6.0      6.0      6.0      1      6.0
0001046934      0.0      1.4E-45  0.0      1      0.0
0001047213      9.0      9.0      9.0      1      9.0
0001047647      0.0      1.4E-45  0.0      1      0.0
0001047663      0.0      1.4E-45  0.0      1      0.0
0001047868      0.0      1.4E-45  0.0      1      0.0
0001047973      9.0      9.0      18.0     2      9.0
000104799X      7.0      8.0      15.0     2      7.5
0001048082      8.0      8.0      16.0     2      8.0
0001048473      0.0      1.4E-45  0.0      1      0.0
0001049879      0.0      1.4E-45  0.0      1      0.0
0001052039      0.0      1.4E-45  0.0      1      0.0
0001053736      5.0      5.0      5.0      1      5.0
0001053744      5.0      5.0      5.0      1      5.0
0001055607      8.0      8.0      8.0      1      8.0
0001056107      8.0      8.0      16.0     2      8.0
0001061127      8.0      8.0      8.0      1      8.0
000123207X      0.0      1.4E-45  0.0      1      0.0
0001232088      0.0      1.4E-45  0.0      1      0.0
0001360469     10.0     10.0     10.0      1     10.0
0001372564      0.0      1.4E-45  0.0      1      0.0
0001374362      9.0      9.0      9.0      1      9.0
0001374869     10.0     10.0     10.0      1     10.0
0001382381      0.0      1.4E-45  0.0      1      0.0
0001382608      0.0      1.4E-45  0.0      1      0.0
000160418X      0.0      7.0      7.0      2      3.5
cat: Unable to write to output stream.
```

2. Top Books: Identify the top books based on the number of ratings/reviews.

This analysis comprises of :

- **Driver Class:** Sets up the configuration for Mapper1, Reducer1, Mapper2 and Reducer2 classes.
- **Mapper 1 Class:** Emits ISBN as the key and 1 as the value for each rating record.
- **Reducer 1 Class:** Counts the total number of ratings/reviews for each ISBN.
- **Mapper 2 Class:** Emits the rating count as the key and ISBN as the value for sorting the books based on the number of ratings.
- **Reducer 2 Class:** Sorts the books in descending order based on the rating count and emits the Top 50 books with their ISBN and rating count.
- **CustomKeyComparator:** Used to sort the books based on the number of ratings in descending order by implementing the Comparable method.

Mapper 1

```
/** Two-stage MapReduce: summarize ratings per ISBN, then globally rank by rating count. */
public class TopBooks {
    public static class Map1 extends Mapper<Object,Text,Text,IntWritable>{
        private final Text isbn=new Text(); private final IntWritable one=new IntWritable(1);
        protected void map(Object key,Text value,Context c)throws IOException,InterruptedException{
            String[] f=CsvUtils.parseLine(value.toString()); if(f.length<7||CsvUtils.isHeader(f))return;
            if(!f[0].trim().isEmpty()){isbn.set(f[0].trim()); c.write(isbn,one);}
        }
    }
}
```

Reducer 1

```
23 public static class Reduce1 extends Reducer<Text,IntWritable,Text,IntWritable>{
24     private final IntWritable out=new IntWritable();
25     protected void reduce(Text k,Iterable<IntWritable> vals,Context c)throws IOException,InterruptedException{int n=0;for(IntWritable v:vals)n+=v.get();out.set(n);c.write(k,out);}
26 }
```

Mapper 2

```
27 public static class Map2 extends Mapper<Object,Text,IntWritable,Text>{
28     protected void map(Object key,Text value,Context c)throws IOException,InterruptedException{
29         String[] r=value.toString().split("\\t"); if(r.length<2)return;
30         try{c.write(new IntWritable(Integer.parseInt(r[1].trim())),new Text(r[0].trim()));}catch(Exception ignored){}
31     }
32 }
```

Reducer 2

```
33 public static class Reduce2 extends Reducer<IntWritable,Text,Text,IntWritable>{
34     private int remaining=50;
35     protected void reduce(IntWritable count,Iterable<Text> books,Context c)throws IOException,InterruptedException{
36         for(Text book:books){if(remaining<=0)return;c.write(book,count);remaining--;}
37     }
38 }
```

Driver

```
39 public static void main(String[] args)throws Exception{
40     if(args.length!=3){System.err.println("Usage: TopBooks <input> <intermediate-output> <final-output>");System.exit(2);}
41     Job j1=Job.getInstance(new Configuration(),"Top Books - Count Ratings"); j1.setJarByClass(TopBooks.class);
42     j1.setMapperClass(Map1.class);j1.setReducerClass(Reduce1.class);j1.setMapOutputKeyClass(Text.class);j1.setMapOutputValueClass(IntWritable.class);j1.setOutputKeyC
43     FileInputFormat.addInputPath(j1,new Path(args[0]));FileOutputFormat.setOutputPath(j1,new Path(args[1]));
44     if(!j1.waitForCompletion(true))System.exit(1);
45     Job j2=Job.getInstance(new Configuration(),"Top 50 Most Rated Books"); j2.setJarByClass(TopBooks.class);
46     j2.setMapperClass(Map2.class);j2.setReducerClass(Reduce2.class);j2.setMapOutputKeyClass(IntWritable.class);j2.setMapOutputValueClass(Text.class);j2.setOutputKeyC
47     FileInputFormat.addInputPath(j2,new Path(args[1]));FileOutputFormat.setOutputPath(j2,new Path(args[2]));
48     System.exit(j2.waitForCompletion(true)?0:1);
49 }
50 }
```

Execution

```
C:\BDLFinalProject\TopBooks>cd C:\BDLFinalProject\build2
C:\BDLFinalProject\build2>jar cf topbooks.jar com
C:\BDLFinalProject\build2>hadoop jar topbooks.jar com.neu.bigdata.TopBooks /BDLFinalProject/data/bookDataset.csv /BDLFinalProject/data/outputFiles/tempCount
/BDLFinalProject/data/outputFiles/top50Books
2026-09-13 10:33:15,559 INFO client.RMProxy: Connecting to ResourceManager at /0.0.0.0:8032
2026-09-13 10:33:16,157 WARN mapreduce.JobResourceUploader: Hadoop command-line option parsing not performed. Implement the Tool interface and execute your
application with ToolRunner to remedy this.
2026-09-13 10:33:16,189 INFO mapreduce.JobResourceUploader: Disabling Erasure Coding for path: /tmp/hadoop-yarn/staging/absar/.staging/job_1789271018238_000
1
2026-09-13 10:33:16,839 INFO input.FileInputFormat: Total input files to process : 1
2026-09-13 10:33:16,923 INFO mapreduce.JobSubmitter: number of splits:1
2026-09-13 10:33:17,078 INFO mapreduce.JobSubmitter: Submitting tokens for job: job_1789271018238_0001
2026-09-13 10:33:17,084 INFO mapreduce.JobSubmitter: Executing with tokens: []
2026-09-13 10:33:17,281 INFO conf.Configuration: resource-types.xml not found
2026-09-13 10:33:17,281 INFO resource.ResourceUtils: Unable to find 'resource-types.xml'.
2026-09-13 10:33:17,752 INFO impl.YarnClientImpl: Submitted application application_1789271018238_0001
2026-09-13 10:33:17,806 INFO mapreduce.Job: The url to track the job: http://DESKTOP-SAJUG07:8088/proxy/application_1789271018238_0001/
2026-09-13 10:33:17,806 INFO mapreduce.Job: Running job: job_1789271018238_0001
2026-09-13 10:33:25,908 INFO mapreduce.Job: Job job_1789271018238_0001 running in uber mode : false
2026-09-13 10:33:25,990 INFO mapreduce.Job:  map 0% reduce 0%
2026-09-13 10:33:34,188 INFO mapreduce.Job:  map 100% reduce 0%
2026-09-13 10:33:42,292 INFO mapreduce.Job:  map 100% reduce 100%
2026-09-13 10:33:42,384 INFO mapreduce.Job: Job job_1789271018238_0001 completed successfully
2026-09-13 10:33:42,420 INFO mapreduce.Job: Counters: 54
File System Counters
  FILE: Number of bytes read=17527409
  FILE: Number of bytes written=35533011
  FILE: Number of read operations=0
  FILE: Number of large read operations=0
  FILE: Number of write operations=0
  HDFS: Number of bytes read=91219055
  HDFS: Number of bytes written=3529593
  HDFS: Number of read operations=0
  HDFS: Number of large read operations=0
  HDFS: Number of write operations=2
  HDFS: Number of bytes read erasure-coded=0
Job Counters
  Completed jobs=1
Total vcore-milliseconds taken by all reduce tasks=3560
Total megabyte-milliseconds taken by all map tasks=3902464
Total megabyte-milliseconds taken by all reduce tasks=3645440
Map-Reduce Framework
  Map input records=270106
  Map output records=270106
  Map output bytes=4051599
  Map output materialized bytes=4591817
  Input split bytes=142
  Combine input records=0
  Combine output records=0
  Reduce input groups=331
  Reduce shuffle bytes=4591817
  Reduce input records=270106
  Reduce output records=50
  Spilled Records=540212
  Shuffled Maps =1
  Failed Shuffles=0
  Merged Map outputs=1
  GC time elapsed (ms)=98
  CPU time spent (ms)=2857
  Physical memory (bytes) snapshot=577400832
  Virtual memory (bytes) snapshot=874827776
  Total committed heap usage (bytes)=495976448
  Peak Map Physical memory (bytes)=353234944
  Peak Map Virtual memory (bytes)=519438336
  Peak Reduce Physical memory (bytes)=224165888
  Peak Reduce Virtual memory (bytes)=360161280
Shuffle Errors
  BAD_ID=0
  CONNECTION=0
  IO_ERROR=0
  WRONG_LENGTH=0
  WRONG_MAP=0
  WRONG_REDUCE=0
File Input Format Counters
  Bytes Read=3529593
File Output Format Counters
  Bytes Written=752
.: \BDLFinalProject\build2>
```

Output

```
C:\BDLFinalProject\build2>hadoop fs -cat /BDLFinalProject/data/outputFiles/top50Books/part-r-00000
0971880107      2502
0316666343      1295
0385504209       883
0060928336       732
0312195516       723
044023722X       647
0142001740       615
067976402X       614
0671027360       586
0446672211       585
059035342X       571
0316601950       568
0375727345       552
044021145X       529
0452282152       526
0440214041       523
0804106304       519
0440211727       517
0345337662       506
0060930535       494
0440226430       482
0312278586       474
0743418174       470
0671021001       468
0345370775       466
0446605239       465
0156027321       462
0440241073       456
0671003755       446
0060976845       434
1400034779       431
0786868716       427
0440234743       422
0440222656       420
0440221471       419
0345361792       412
0440236673       408
0345417623       407
0316769487       403
```

```
0316769487      403
0446610038      391
0385484518      391
0446310786      389
044022165X      386
0375706771      380
0440225701      379
0440220602      379
0060502258      376
0446606812      369
0345353145      368
044651652X      368
```

3. Top Books by Publisher: Identify the top books for each publisher based on the number of ratings/reviews.

This analysis comprises of:

- **Driver Class:** Sets up the configuration for Mapper1, Reducer1, Mapper2 and Reducer2 classes.
- **Mapper 1 Class:** Emits ISBN as the key and Publisher and 1 as the value for each rating record.
- **Reducer 1 Class:** Groups records by ISBN and Publisher and calculates the total number of ratings/reviews for each book.
- **Mapper 2 Class:** Emits the Publisher and rating count as the key and ISBN as the value to group and sort books according to their rating counts within each publisher.
- **Reducer 2 Class:** Sorts the books for each publisher in descending order based on the number of ratings/reviews and emits the Top 25 books for each publisher.

Mapper 1

```
/** Finds the top 25 books by number of ratings inside each publisher. */
public class TopBooksByPublisher {
    public static class Map1 extends Mapper<Object,Text,Text,Text>{
        private final Text publisher=new Text(), book=new Text();
        protected void map(Object k,Text v,Context c)throws IOException,InterruptedException{String[] f=CsvUtils.parseLine(v.toString());
        if(f.length<7||CsvUtils.isHeader(f))return;
        String p=f[4].trim(),isbn=f[0].trim();
        if(!p.isEmpty()&&!isbn.isEmpty()){publisher.set(p);
        book.set(isbn);
        c.write(publisher,book);}}
    }
}
```

Reducer 1

```
public static class Reduce1 extends Reducer<Text,Text,Text,Text>{
    protected void reduce(Text p,Iterable<Text> books,Context c)throws IOException,InterruptedException{Map<String,Integer> counts=new HashMap<String,Integer>();
    for(Text b:books){String s=b.toString();
    Integer n=counts.get(s);
    counts.put(s,n==null?1:n+1);
    }
    for(Map.Entry<String,Integer> e:counts.entrySet())c.write(new Text(e.getKey()),new Text(p.toString()+"\t"+e.getValue()));}
}
```

Mapper 2

```
public static class Map2 extends Mapper<Object,Text,PublisherCountKey,Text>{
    protected void map(Object k,Text v,Context c)throws IOException,InterruptedException{String[] r=v.toString().split("\\t");
    if(r.length<3)return;try{c.write(new PublisherCountKey(r[1],Integer.parseInt(r[2])),new Text(r[0]));}catch(Exception ignored){}}
}
```

Reducer 2

```
public static class Reduce2 extends Reducer<PublisherCountKey,Text,Text,IntWritable>{
    private String current="";
    private int n=0;
    protected void reduce(PublisherCountKey k,Iterable<Text> books,Context c)throws IOException,InterruptedException{if(!k.publisher.equals(current)){current=k.publisher;
    n=0;}
    for(Text b:books){if(n>=25)return;c.write(new Text(current+"\t"+b.toString()),new IntWritable(k.count));
    n++;}
}
```

Driver

```
public static void main(String[] a)throws Exception{if(a.length!=3)
{System.err.println("Usage: TopBooksByPublisher <input> <intermediate-output> <final-output>");
System.exit(2);}Job j1=Job.getInstance(new Configuration(),"Top Books By Publisher - Count");
j1.setJarByClass(TopBooksByPublisher.class);j1.setMapperClass(Map1.class);
j1.setReducerClass(Reduce1.class);
j1.setMapOutputKeyClass(Text.class);
j1.setMapOutputValueClass(Text.class);
j1.setOutputKeyClass(Text.class);
j1.setOutputValueClass(Text.class);
FileInputFormat.addInputPath(j1,new Path(a[0]));
FileOutputFormat.setOutputPath(j1,new Path(a[1]));
if(!j1.waitForCompletion(true))System.exit(1);
Job j2=Job.getInstance(new Configuration(),"Top 25 Books By Publisher");
j2.setJarByClass(TopBooksByPublisher.class);j2.setMapperClass(Map2.class);
j2.setReducerClass(Reduce2.class);j2.setMapOutputKeyClass(PublisherCountKey.class);
j2.setMapOutputValueClass(Text.class);j2.setOutputKeyClass(Text.class);
j2.setOutputValueClass(IntWritable.class);j2.setNumReduceTasks(1);
FileInputFormat.addInputPath(j2,new Path(a[1]));
FileOutputFormat.setOutputPath(j2,new Path(a[2]));
System.exit(j2.waitForCompletion(true)?0:1);
}
```

Execution

```
C:\BDLFinalProject\TopBooksByPublisher>cd C:\BDLFinalProject\build3
C:\BDLFinalProject\build3>jar cf topbookspub.jar com
C:\BDLFinalProject\build3>hadoop jar topbookspub.jar com.neu.bigdata.TopBooksByPublisher /BDLFinalProject/data/bookDataset.csv /BDLFinalProject/data/outputFiles/tempPub /BDLFinalProject/data/outputFiles/top25ByPublisher
2026-09-13 10:46:08,366 INFO client.RMPProxy: Connecting to ResourceManager at /0.0.0.0:8032
2026-09-13 10:46:08,947 WARN mapreduce.JobResourceUploader: Hadoop command-line option parsing not performed. Implement the Tool interface and execute your application with ToolRunner to remedy this.
2026-09-13 10:46:08,952 INFO mapreduce.JobResourceUploader: Disabling Erasure Coding for path: /tmp/hadoop-yarn/staging/absar/.staging/job_1789271018238_0003
2026-09-13 10:46:09,111 INFO input.FileInputFormat: Total input files to process : 1
2026-09-13 10:46:09,175 INFO mapreduce.JobSubmitter: number of splits:1
2026-09-13 10:46:09,289 INFO mapreduce.JobSubmitter: Submitting tokens for job: job_1789271018238_0003
2026-09-13 10:46:09,291 INFO mapreduce.JobSubmitter: Executing with tokens: []
2026-09-13 10:46:09,420 INFO conf.Configuration: resource-types.xml not found
2026-09-13 10:46:09,420 INFO resource.ResourceUtils: Unable to find 'resource-types.xml'.
2026-09-13 10:46:09,467 INFO impl.YarnClientImpl: Submitted application application_1789271018238_0003
2026-09-13 10:46:09,493 INFO mapreduce.Job: The url to track the job: http://DESKTOP-SAJUGD7:8088/proxy/application_1789271018238_0003/
2026-09-13 10:46:09,493 INFO mapreduce.Job: Running job: job_1789271018238_0003
2026-09-13 10:46:16,670 INFO mapreduce.Job: Job job_1789271018238_0003 running in uber mode : false
2026-09-13 10:46:16,670 INFO mapreduce.Job: map 0% reduce 0%
2026-09-13 10:46:24,804 INFO mapreduce.Job: map 100% reduce 0%
2026-09-13 10:46:31,900 INFO mapreduce.Job: map 100% reduce 100%
2026-09-13 10:46:32,914 INFO mapreduce.Job: Job job_1789271018238_0003 completed successfully
2026-09-13 10:46:33,001 INFO mapreduce.Job: Counters: 54
  File System Counters
    FILE: Number of bytes read=29123048
    FILE: Number of bytes written=58724311
    FILE: Number of read operations=0
    FILE: Number of large read operations=0
    FILE: Number of write operations=0
    HDFS: Number of bytes read=91219855
    HDFS: Number of bytes written=8099444
    HDFS: Number of read operations=8
    HDFS: Number of large read operations=0
    HDFS: Number of write operations=2
    HDFS: Number of bytes read erasure-coded=0
  Job Counters
    Launched map tasks=1
```

```
Total vcore-milliseconds taken by all reduce tasks=4315
Total megabyte-milliseconds taken by all map tasks=4886528
Total megabyte-milliseconds taken by all reduce tasks=4418560
Map-Reduce Framework
  Map input records=270104
  Map output records=270104
  Map output bytes=8891550
  Map output materialized bytes=9431766
  Input split bytes=140
  Combine input records=0
  Combine output records=0
  Reduce input groups=35699
  Reduce shuffle bytes=9431766
  Reduce input records=270104
  Reduce output records=76928
  Spilled Records=540208
  Shuffled Maps =1
  Failed Shuffles=0
  Merged Map outputs=1
  GC time elapsed (ms)=120
  CPU time spent (ms)=5590
  Physical memory (bytes) snapshot=718221312
  Virtual memory (bytes) snapshot=995360768
  Total committed heap usage (bytes)=664797184
  Peak Map Physical memory (bytes)=475623424
  Peak Map Virtual memory (bytes)=610271232
  Peak Reduce Physical memory (bytes)=242597888
  Peak Reduce Virtual memory (bytes)=385183744
Shuffle Errors
  BAD_ID=0
  CONNECTION=0
  IO_ERROR=0
  WRONG_LENGTH=0
  WRONG_MAP=0
  WRONG_REDUCE=0
File Input Format Counters
  Bytes Read=8099444
File Output Format Counters
  Bytes Written=2466736
```

Output

```
C:\BDLFinalProject\build3>hadoop fs -cat /BDLFinalProject/data/outputFiles/top25ByPublisher/part-r-00000 | more
'K' Pub 9838523283 1
(3 Queen Sq., WC1N 3AU) Faber and Faber Ltd 0571097006 1
(49 Poland St., W.1) BPC Unit 75 0356037290 1
10-18 2264003790 8
10-18 226401699X 6
10-18 2264019638 3
10-18 2264017627 3
10-18 2264017635 3
10-18 2264018917 3
10-18 2264017295 3
10-18 2264005432 3
10-18 2264001879 3
10-18 2264017317 2
10-18 2264019646 2
10-18 2264026545 2
10-18 2264005785 2
10-18 2264017805 2
10-18 2264015667 2
10-18 2264025859 2
10-18 2264001445 1
10-18 2264019654 1
10-18 2264018542 1
10-18 2264016981 1
10-18 2264018941 1
10-18 2264016523 1
10-18 2264006609 1
10-18 2264011092 1
10-18 226400598X 1
10/18 2264032804 3
10/18 2264035854 2
10/18 226403551X 2
10/18 226403629X 2
10/18 2264034157 1
10/18 2264034955 1
10/18 2264033894 1
10/18 2264034823 1
10/18 2264035528 1
10/18 2264034831 1
10/18 2264035498 1
10/18 2264031808 1
```

Pig YouTube Data Analysis

1. Calculate the Top 5 Publishers based on the number of distinct books: Returns the Top 5 publishers based on the number of distinct books that have received rating

Script

```
-- Top 5 publishers by number of distinct rated books
books = LOAD 'hdfs:///BDLFinalProject/data/bookDataset.csv' USING PigStorage(',') AS
(isbn:chararray, title:chararray, author:chararray, year:int, publisher:chararray,
userid:chararray, rating:float);
filtered = FILTER books BY isbn IS NOT NULL AND isbn != 'ISBN' AND publisher IS NOT NULL
AND publisher != '';
grp = GROUP filtered BY publisher;
counts = FOREACH grp { distinct_books = DISTINCT filtered.isbn; GENERATE group AS
publisher, COUNT(distinct_books) AS book_count; };
sorted = ORDER counts BY book_count DESC;
top5 = LIMIT sorted 5;
STORE top5 INTO 'Top5Publishers.txt' using PigStorage('|');
```

Execution

```
C:\BDLFinalProject\PigAnalysis\scripts>pig -x mapreduce top5Publishers.pig
2026-09-13 12:58:15,410 INFO pig.ExecTypeProvider: Trying ExecType : LOCAL
2026-09-13 12:58:15,413 INFO pig.ExecTypeProvider: Trying ExecType : MAPREDUCE
2026-09-13 12:58:15,415 INFO pig.ExecTypeProvider: Picked MAPREDUCE as the ExecType
2026-09-13 12:58:15,589 [main] INFO org.apache.pig.Main - Apache Pig version 0.18.0 (r1926484) compiled Jun 16 2025, 10:38:21
2026-09-13 12:58:15,589 [main] INFO org.apache.pig.Main - Logging error messages to: C:\hadoop\logs\pig.1789282695589.log
2026-09-13 12:58:15,969 [main] INFO org.apache.pig.impl.util.Utils - Default bootstrap file C:\Users\absar/.pigbootstrap not found
2026-09-13 12:58:16,024 [main] INFO org.apache.hadoop.conf.Configuration.deprecation - mapred.job.tracker is deprecated. Instead, use mapreduce.jobtracker.address
2026-09-13 12:58:16,024 [main] INFO org.apache.pig.backend.hadoop.executionengine.HExecutionEngine - Connecting to hadoop file system at: hdfs://localhost:9000
2026-09-13 12:58:16,364 [main] INFO org.apache.pig.PigServer - Pig Script ID for the session: PIG-top5Publishers.pig-0403fafd-4982-4361-8a1b-105832698dc6
2026-09-13 12:58:16,364 [main] WARN org.apache.pig.PigServer - ATS is disabled since yarn.timeline-service.enabled set to false
2026-09-13 12:58:16,881 [main] INFO org.apache.hadoop.conf.Configuration.deprecation - mapred.textoutputformat.separator is deprecated. Instead, use mapreduce.textoutputformat.separator
2026-09-13 12:58:16,902 [main] INFO org.apache.pig.tools.pigstats.ScriptState - Pig features used in the script: GROUP_BY,ORDER_BY,FILTER,LIMIT
2026-09-13 12:58:16,918 [main] INFO org.apache.pig.data.SchemaTupleBackend - Key [pig.schematuple] was not set... will not generate code.
2026-09-13 12:58:16,933 [main] INFO org.apache.pig.newplan.logical.optimizer.LogicalPlanOptimizer - {RULES_ENABLED=[AddForEach, ColumnMapKeyPrune, FilterConstantCalculator, ForEachConstantCalculator, GroupByConstParallelSetter, LimitOptimizer, LoadTypeCastInserter, MergeFilter, MergeForEach, NestedLimitOptimizer, PartitionFilterOptimizer, PredicatePushdownOptimizer, PushDownForEachFlatten, PushUpFilter, SplitConstantCalculator, SplitFilter, StreamTypeCastInserter]}
2026-09-13 12:58:16,989 [main] INFO org.apache.pig.impl.util.SpillableMemoryManager - Selected heap (PS Old Gen) of size 699400192 to monitor. collectionUsageThreshold = 489580128, usageThreshold = 489580128
2026-09-13 12:58:17,031 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MRCCompiler - File concatenation threshold: 100 optimistic? false
2026-09-13 12:58:17,099 [main] INFO org.apache.pig.backend.hadoop.executionengine.util.CombinerOptimizerUtil - Choosing to move algebraic foreach to combiner
2026-09-13 12:58:17,109 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.SecondaryKeyOptimizerMR - Using Secondary Key Optimization for MapReduce node scope=59
2026-09-13 12:58:17,118 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MultiQueryOptimizer - MR plan size before optimization: 4
2026-09-13 12:58:17,118 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MultiQueryOptimizer - MR plan size after optimization: 4
2026-09-13 12:58:17,181 [main] INFO org.apache.hadoop.yarn.client.RMProxy - Connecting to ResourceManager at /0.0.0.0:8032
2026-09-13 12:58:17,331 [main] INFO org.apache.hadoop.conf.Configuration.deprecation - yarn.resourcemanager.system-metrics-publisher.enabled is deprecated. Instead, use yarn.system-metrics-publisher.enabled
2026-09-13 12:58:17,347 [main] INFO org.apache.pig.tools.pigstats.mapreduce.MRScriptState - Pig script settings are added to the job
2026-09-13 12:58:17,359 [main] INFO org.apache.hadoop.conf.Configuration.deprecation - mapred.job.reduce.markreset.buffer.percent is deprecated. Instead, use mapreduce.reduce.markreset.buffer.percent
2026-09-13 12:58:17,361 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.JobControlCompiler - mapred.job.reduce.markreset.buffer.percent is not set, set to default 0.3
2026-09-13 12:58:17,363 [main] INFO org.apache.hadoop.conf.Configuration.deprecation - mapred.output.compress is deprecated. Instead, use mapreduce.output.compressor

2026-09-13 13:14:22,446 [main] INFO org.apache.pig.data.SchemaTupleFrontend - Setting key [pig.schematuple.classes] with classes to deserialize []
2026-09-13 13:14:22,460 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MapReduceLauncher - 1 map-reduce job(s) waiting for submission.
2026-09-13 13:14:22,466 [JobControl] INFO org.apache.hadoop.yarn.client.RMProxy - Connecting to ResourceManager at /0.0.0.0:8032
2026-09-13 13:14:22,500 [JobControl] INFO org.apache.hadoop.mapreduce.JobResourceUploader - Disabling Erasure Coding for path: /tmp/hadoop-yarn/staging/absar/.staging/job.1789271018238_0025
2026-09-13 13:14:22,504 [JobControl] WARN org.apache.hadoop.mapreduce.JobResourceUploader - No job jar file set. User classes may not be found. See Job or Job#setJar(String).
2026-09-13 13:14:22,540 [JobControl] INFO org.apache.hadoop.mapreduce.lib.input.FileInputFormat - Total input files to process : 1
2026-09-13 13:14:22,542 [JobControl] INFO org.apache.pig.backend.hadoop.executionengine.util.MapRedUtil - Total input paths to process : 1
2026-09-13 13:14:22,542 [JobControl] INFO org.apache.pig.backend.hadoop.executionengine.util.MapRedUtil - Total input paths (combined) to process : 1
2026-09-13 13:14:22,634 [JobControl] INFO org.apache.hadoop.mapreduce.JobSubmitter - number of splits:1
2026-09-13 13:14:22,662 [JobControl] INFO org.apache.hadoop.mapreduce.JobSubmitter - Submitting tokens for job: job_1789271018238_0025
2026-09-13 13:14:22,665 [JobControl] INFO org.apache.hadoop.mapreduce.JobSubmitter - Executing with tokens: []
2026-09-13 13:14:22,669 [JobControl] INFO org.apache.hadoop.mapred.YARNRunner - Job jar is not present. Not adding any jar to the list of resources.
2026-09-13 13:14:22,905 [JobControl] INFO org.apache.hadoop.yarn.client.api.impl.YarnClientImpl - Submitted application application_1789271018238_0025
2026-09-13 13:14:22,915 [JobControl] INFO org.apache.hadoop.mapreduce.Job - The url to track the job: http://DESKTOP-SAJUGD7:8088/proxy/application_1789271018238_0025/
2026-09-13 13:14:22,969 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MapReduceLauncher - HadoopJobId: job_1789271018238_0025
2026-09-13 13:14:22,970 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MapReduceLauncher - Processing aliases sorted
2026-09-13 13:14:22,970 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MapReduceLauncher - detailed locations: M: sorted[6,9] C: R:
2026-09-13 13:14:40,274 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MapReduceLauncher - 87% complete
2026-09-13 13:14:40,276 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MapReduceLauncher - Running jobs are [job_1789271018238_0025]
2026-09-13 13:14:50,521 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MapReduceLauncher - Running jobs are [job_1789271018238_0025]
2026-09-13 13:14:53,084 [main] INFO org.apache.hadoop.yarn.client.RMProxy - Connecting to ResourceManager at /0.0.0.0:8032
2026-09-13 13:14:53,096 [main] INFO org.apache.hadoop.mapred.ClientServiceDelegate - Application state is completed. FinalApplicationStatus=SUCCEEDED. Redirecting to job history server
2026-09-13 13:14:56,132 [main] INFO org.apache.hadoop.ipc.Client - Retrying connect to server: 0.0.0.0/0.0.0.0:10020. Already tried 0 time(s); retry policy is RetryUpToMaximumCountWithFixedSleep(maxRetries=10, sleepTime=1000 MILLISECONDS)
2026-09-13 13:14:59,182 [main] INFO org.apache.hadoop.ipc.Client - Retrying connect to server: 0.0.0.0/0.0.0.0:10020. Already tried 1 time(s); retry policy is RetryUpToMaximumCountWithFixedSleep(maxRetries=10, sleepTime=1000 MILLISECONDS)
2026-09-13 13:15:02,238 [main] INFO org.apache.hadoop.ipc.Client - Retrying connect to server: 0.0.0.0/0.0.0.0:10020. Already tried 2 time(s); retry policy is RetryUpToMaximumCountWithFixedSleep(maxRetries=10, sleepTime=1000 MILLISECONDS)
2026-09-13 13:15:05,291 [main] INFO org.apache.hadoop.ipc.Client - Retrying connect to server: 0.0.0.0/0.0.0.0:10020. Already tried 3 time(s); retry policy is RetryUpToMaximumCountWithFixedSleep(maxRetries=10, sleepTime=1000 MILLISECONDS)
Terminate batch job (Y/N)? N
```

Output

```
C:\BDLFinalProject\PigAnalysis\scripts>hadoop fs -cat Top5Publishers.txt/part-r-00000
Harlequin|7524
Silhouette|4190
Pocket|3896
Ballantine Books|3775
Bantam Books|3640
```

2. Calculate the Top 10 Most Reviewed Books: Returns the Top 10 most-reviewed books based on the total number of ratings/reviews received.

Script

```
-- Top 10 books with the highest number of ratings/reviews
books = LOAD 'hdfs:///BDLFinalProject/data/bookDataset.csv' USING PigStorage(',') AS
(isbn:chararray, title:chararray, author:chararray, year:int, publisher:chararray,
userid:chararray, rating:float);
filtered = FILTER books BY isbn IS NOT NULL AND isbn != 'ISBN';
grp = GROUP filtered BY isbn;
summary = FOREACH grp {
    one = LIMIT filtered 1;
    GENERATE group AS isbn, FLATTEN(one.title) AS title, COUNT(filtered) AS review_count;
};
sorted = ORDER summary BY review_count DESC;
top10 = LIMIT sorted 10;
STORE top10 INTO 'Top10MostReviewed.txt' using PigStorage('|');
```

Execution

```
C:\BDLFinalProject\PigAnalysis\scripts>pig -x mapreduce top10MostReviewed.pig
2026-09-13 12:12:02,609 INFO pig.ExecTypeProvider: Trying ExecType : LOCAL
2026-09-13 12:12:02,616 INFO pig.ExecTypeProvider: Trying ExecType : MAPREDUCE
2026-09-13 12:12:02,616 INFO pig.ExecTypeProvider: Picked MAPREDUCE as the ExecType
2026-09-13 12:12:02,798 [main] INFO org.apache.pig.Main - Apache Pig version 0.18.0 (r1926484) compiled Jun 16 2025, 10:38:21
2026-09-13 12:12:02,798 [main] INFO org.apache.pig.Main - Logging error messages to: C:\hadoop\logs\pig_1789279922798.log
2026-09-13 12:12:03,203 [main] INFO org.apache.pig.impl.util.Utils - Default bootstrap file C:\Users\absar/.pigbootstrap not found
2026-09-13 12:12:03,246 [main] INFO org.apache.hadoop.conf.Configuration.deprecation - mapred.job.tracker is deprecated. Instead, use mapreduce.jobtracker.address
2026-09-13 12:12:03,246 [main] INFO org.apache.pig.backend.hadoop.executionengine.MExecutionEngine - Connecting to hadoop file system at: hdfs://localhost:9000
2026-09-13 12:12:03,603 [main] INFO org.apache.pig.PigServer - Pig Script ID for the session: PIG-top10MostReviewed.pig-51874095-7189-466e-a16a-b12a61680f70
2026-09-13 12:12:03,603 [main] WARN org.apache.pig.PigServer - ATS is disabled since yarn.timeline-service.enabled set to false
2026-09-13 12:12:04,136 [main] INFO org.apache.hadoop.conf.Configuration.deprecation - mapred.textoutputformat.separator is deprecated. Instead, use mapreduce.output.textoutputformat.separator
2026-09-13 12:12:04,149 [main] INFO org.apache.pig.tools.pigstats.ScriptState - Pig features used in the script: GROUP_BY,ORDER_BY,FILTER,LIMIT
2026-09-13 12:12:04,173 [main] INFO org.apache.pig.data.SchemaTupleBackend - Key [pig.schematuple] was not set... will not generate code.
2026-09-13 12:12:04,191 [main] INFO org.apache.pig.newplan.logical.optimizer.LogicalPlanOptimizer - {RULES_ENABLED=[AddForEach, ColumnMapKeyPrune, FilterConstantCalculator, ForEachConstantCalculator, GroupByConstParallelSetter, LimitOptimizer, LoadTypeCastInserter, MergeFilter, MergeForEach, NestedLimitOptimizer, PartitionFilterOptimizer, PredicatePushdownOptimizer, PushDownForEachFlatten, PushUpFilter, SplitConstantCalculator, SplitFilter, StreamTypeCastInserter]}
2026-09-13 12:12:04,235 [main] INFO org.apache.pig.impl.util.SpillableMemoryManager - Selected heap (PS Old Gen) of size 699400192 to monitor. collectionUsageThreshold = 489580128, usageThreshold = 489580128
2026-09-13 12:12:04,279 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MRCompiler - File concatenation threshold: 100 optimistic? false
2026-09-13 12:12:04,353 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.SecondaryKeyOptimizerMR - Using Secondary Key Optimizator for MapReduce node scope-51
2026-09-13 12:12:04,353 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MultiQueryOptimizer - MR plan size before optimization: 4
2026-09-13 12:12:04,353 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MultiQueryOptimizer - MR plan size after optimization: 4
2026-09-13 12:12:04,419 [main] INFO org.apache.hadoop.yarn.client.RMProxy - Connecting to ResourceManager at /0.0.0.0:8032
2026-09-13 12:12:04,580 [main] INFO org.apache.hadoop.conf.Configuration.deprecation - yarn.resourcemanager.system-metrics-publisher.enabled is deprecated. Instead, use yarn.system-metrics-publisher.enabled
2026-09-13 12:12:04,596 [main] INFO org.apache.pig.tools.pigstats.mapreduce.MRScriptState - Pig script settings are added to the job
2026-09-13 12:12:04,604 [main] INFO org.apache.hadoop.conf.Configuration.deprecation - mapred.job.reduce.markreset.buffer.percent is deprecated. Instead, use mapreduce.reduce.markreset.buffer.percent
2026-09-13 12:12:04,606 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.JobControlCompiler - mapred.job.reduce.markreset.buffer.percent is not set, set to default 0.3
2026-09-13 12:12:04,611 [main] INFO org.apache.hadoop.conf.Configuration.deprecation - mapred.output.compress is deprecated. Instead, use mapreduce.output.textoutputformat.compress
2026-09-13 12:23:52,811 [main] INFO org.apache.hadoop.ipc.Client - Retrying connect to server: 0.0.0.0/0.0.0.0:10020. Already tried 8 time(s); retry policy is RetryUpToMaximumCountWithFixedSleep(maxRetries=10, sleepTime=1000 MILLISECONDS)
2026-09-13 12:23:55,853 [main] INFO org.apache.hadoop.ipc.Client - Retrying connect to server: 0.0.0.0/0.0.0.0:10020. Already tried 9 time(s); retry policy is RetryUpToMaximumCountWithFixedSleep(maxRetries=10, sleepTime=1000 MILLISECONDS)
2026-09-13 12:23:58,011 [main] INFO org.apache.hadoop.mapred.ClientServiceDelegate - Application state is completed. FinalApplicationStatus=SUCCEEDED. Redirecting to job history server
2026-09-13 12:24:01,057 [main] INFO org.apache.hadoop.ipc.Client - Retrying connect to server: 0.0.0.0/0.0.0.0:10020. Already tried 0 time(s); retry policy is RetryUpToMaximumCountWithFixedSleep(maxRetries=10, sleepTime=1000 MILLISECONDS)
2026-09-13 12:24:04,100 [main] INFO org.apache.hadoop.ipc.Client - Retrying connect to server: 0.0.0.0/0.0.0.0:10020. Already tried 1 time(s); retry policy is RetryUpToMaximumCountWithFixedSleep(maxRetries=10, sleepTime=1000 MILLISECONDS)
2026-09-13 12:24:07,142 [main] INFO org.apache.hadoop.ipc.Client - Retrying connect to server: 0.0.0.0/0.0.0.0:10020. Already tried 2 time(s); retry policy is RetryUpToMaximumCountWithFixedSleep(maxRetries=10, sleepTime=1000 MILLISECONDS)
2026-09-13 12:24:10,189 [main] INFO org.apache.hadoop.ipc.Client - Retrying connect to server: 0.0.0.0/0.0.0.0:10020. Already tried 3 time(s); retry policy is RetryUpToMaximumCountWithFixedSleep(maxRetries=10, sleepTime=1000 MILLISECONDS)
2026-09-13 12:24:13,243 [main] INFO org.apache.hadoop.ipc.Client - Retrying connect to server: 0.0.0.0/0.0.0.0:10020. Already tried 4 time(s); retry policy is RetryUpToMaximumCountWithFixedSleep(maxRetries=10, sleepTime=1000 MILLISECONDS)
2026-09-13 12:24:16,289 [main] INFO org.apache.hadoop.ipc.Client - Retrying connect to server: 0.0.0.0/0.0.0.0:10020. Already tried 5 time(s); retry policy is RetryUpToMaximumCountWithFixedSleep(maxRetries=10, sleepTime=1000 MILLISECONDS)
2026-09-13 12:24:19,335 [main] INFO org.apache.hadoop.ipc.Client - Retrying connect to server: 0.0.0.0/0.0.0.0:10020. Already tried 6 time(s); retry policy is RetryUpToMaximumCountWithFixedSleep(maxRetries=10, sleepTime=1000 MILLISECONDS)
2026-09-13 12:24:22,387 [main] INFO org.apache.hadoop.ipc.Client - Retrying connect to server: 0.0.0.0/0.0.0.0:10020. Already tried 7 time(s); retry policy is RetryUpToMaximumCountWithFixedSleep(maxRetries=10, sleepTime=1000 MILLISECONDS)
2026-09-13 12:24:25,437 [main] INFO org.apache.hadoop.ipc.Client - Retrying connect to server: 0.0.0.0/0.0.0.0:10020. Already tried 8 time(s); retry policy is RetryUpToMaximumCountWithFixedSleep(maxRetries=10, sleepTime=1000 MILLISECONDS)
2026-09-13 12:24:28,489 [main] INFO org.apache.hadoop.ipc.Client - Retrying connect to server: 0.0.0.0/0.0.0.0:10020. Already tried 9 time(s); retry policy is RetryUpToMaximumCountWithFixedSleep(maxRetries=10, sleepTime=1000 MILLISECONDS)
2026-09-13 12:24:30,644 [main] INFO org.apache.hadoop.mapred.ClientServiceDelegate - Application state is completed. FinalApplicationStatus=SUCCEEDED. Redirecting to job history server
2026-09-13 12:24:33,697 [main] INFO org.apache.hadoop.ipc.Client - Retrying connect to server: 0.0.0.0/0.0.0.0:10020. Already tried 0 time(s); retry policy is RetryUpToMaximumCountWithFixedSleep(maxRetries=10, sleepTime=1000 MILLISECONDS)
2026-09-13 12:24:36,751 [main] INFO org.apache.hadoop.ipc.Client - Retrying connect to server: 0.0.0.0/0.0.0.0:10020. Already tried 1 time(s); retry policy is RetryUpToMaximumCountWithFixedSleep(maxRetries=10, sleepTime=1000 MILLISECONDS)
2026-09-13 12:24:39,800 [main] INFO org.apache.hadoop.ipc.Client - Retrying connect to server: 0.0.0.0/0.0.0.0:10020. Already tried 2 time(s); retry policy is RetryUpToMaximumCountWithFixedSleep(maxRetries=10, sleepTime=1000 MILLISECONDS)
2026-09-13 12:24:42,852 [main] INFO org.apache.hadoop.ipc.Client - Retrying connect to server: 0.0.0.0/0.0.0.0:10020. Already tried 3 time(s); retry policy is RetryUpToMaximumCountWithFixedSleep(maxRetries=10, sleepTime=1000 MILLISECONDS)
2026-09-13 12:24:45,910 [main] INFO org.apache.hadoop.ipc.Client - Retrying connect to server: 0.0.0.0/0.0.0.0:10020. Already tried 4 time(s); retry policy is RetryUpToMaximumCountWithFixedSleep(maxRetries=10, sleepTime=1000 MILLISECONDS)
Terminate batch job (Y/N)? N
```

Output

```
C:\BDLFinalProject\PigAnalysis\scripts>hadoop fs -cat Top10MostReviewed.txt/part-r-00000
0971880107|Wild Animus|2502
0316666343|The Lovely Bones: A Novel|1295
0385504209|The Da Vinci Code|883
0060928336|Divine Secrets of the Ya-Ya Sisterhood: A Novel|732
0312195516|The Red Tent (Bestselling Backlist)|723
044023722X|A Painted House|647
0142001740|The Secret Life of Bees|615
067976402X|Snow Falling on Cedars|614
0671027360|Angels & Demons|586
0446672211|Where the Heart Is (Oprah's Book Club (Paperback))|585
```

3. Calculate the Top 10 Most Reviewed Books by Publisher: Returns the Top 10 most-reviewed books for each publisher based on the number of ratings/reviews received.

Script

```
-- Top 10 most-reviewed books inside each publisher
books = LOAD 'hdfs:///BDLFinalProject/data/bookDataset.csv' USING PigStorage(',') AS
(isbn:chararray, title:chararray, author:chararray, year:int, publisher:chararray,
userid:chararray, rating:float);
filtered = FILTER books BY isbn IS NOT NULL AND isbn != 'ISBN' AND publisher IS NOT NULL
AND publisher != '';
bookgrp = GROUP filtered BY (publisher,isbn);
booksummary = FOREACH bookgrp {
    one = LIMIT filtered 1;
    GENERATE group.publisher AS publisher, group.isbn AS isbn, FLATTEN(one.title) AS
title, COUNT(filtered) AS review_count;
};
pubgrp = GROUP booksummary BY publisher;
top_each = FOREACH pubgrp {
    ordered = ORDER booksummary BY review_count DESC;
    limited = LIMIT ordered 10;
    GENERATE group AS publisher, FLATTEN(limited);
};
STORE top_each INTO 'Top10MostReviewedByPublisher.txt' using PigStorage('|');
```

Execution

```
C:\BDLFinalProject\PigAnalysis\scripts>pig -x mapreduce top10MostReviewedByPublisher.pig
2026-09-13 12:49:09,884 INFO pig.ExecTypeProvider: Trying ExecType : LOCAL
2026-09-13 12:49:09,884 INFO pig.ExecTypeProvider: Trying ExecType : MAPREDUCE
2026-09-13 12:49:09,886 INFO pig.ExecTypeProvider: Picked MAPREDUCE as the ExecType
2026-09-13 12:49:10,065 [main] INFO org.apache.pig.Main - Apache Pig version 0.18.0 (r1926484) compiled Jun 16 2025, 10:38:21
2026-09-13 12:49:10,065 [main] INFO org.apache.pig.Main - Logging error messages to: C:\hadoop\logs\pig_1789282150061.log
2026-09-13 12:49:10,425 [main] INFO org.apache.pig.impl.util.Utils - Default bootup file C:\Users\absar/.pigbootup not found
2026-09-13 12:49:10,475 [main] INFO org.apache.hadoop.conf.Configuration.deprecation - mapred.job.tracker is deprecated. Instead, use mapreduce.jobtracker.address
2026-09-13 12:49:10,475 [main] INFO org.apache.pig.backend.hadoop.executionengine.HExecutionEngine - Connecting to hadoop file system at: hdfs://localhost:9000
2026-09-13 12:49:10,844 [main] INFO org.apache.pig.PigServer - Pig Script ID for the session: PIG-top10MostReviewedByPublisher.pig-1004bb69-301d-402b-8989-45c1d569f7de
2026-09-13 12:49:10,844 [main] WARN org.apache.pig.PigServer - ATS is disabled since yarn.timeline-service.enabled set to false
2026-09-13 12:49:11,375 [main] INFO org.apache.hadoop.conf.Configuration.deprecation - mapred.textoutputformat.separator is deprecated. Instead, use mapreduce.output.textoutputformat.separator
2026-09-13 12:49:11,375 [main] INFO org.apache.pig.tools.pigstats.ScriptState - Pig features used in the script: GROUP_BY_FILTER
2026-09-13 12:49:11,392 [main] INFO org.apache.pig.data.SchemaTupleBackend - Key [pig.schematuple] was not set... will not generate code.
2026-09-13 12:49:11,409 [main] INFO org.apache.pig.newplan.logical.optimizer.LogicalPlanOptimizer - {RULES_ENABLED=[AddForEach, ColumnMapKeyPrune, FilterConstantCalculator, ForEachConstantCalculator, GroupByConstParallelSetter, LimitOptimizer, LoadTypeCastInserter, MergeFilter, MergeForEach, NestedLimitOptimizer, PartitionFilterOptimizer, PredicatePushdownOptimizer, PushDownForEachFlatten, PushUpFilter, SplitConstantCalculator, SplitFilter, StreamTypeCastInserter]}
2026-09-13 12:49:11,457 [main] INFO org.apache.pig.impl.util.SpillableMemoryManager - Selected heap (PS Old Gen) of size 699400192 to monitor. collectionUsageThreshold = 489580128, usageThreshold = 489580128
2026-09-13 12:49:11,490 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MRCompiler - File concatenation threshold: 100 optimistic? false
2026-09-13 12:49:11,535 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MultiQueryOptimizer - MR plan size before optimization: 2
2026-09-13 12:49:11,535 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MultiQueryOptimizer - MR plan size after optimization: 2
2026-09-13 12:49:11,593 [main] INFO org.apache.hadoop.yarn.client.RMProxy - Connecting to ResourceManager at /0.0.0.0:8032
2026-09-13 12:49:11,747 [main] INFO org.apache.hadoop.conf.Configuration.deprecation - yarn.resourcemanager.system-metrics-publisher.enabled is deprecated. Instead, use yarn.system-metrics-publisher.enabled
2026-09-13 12:49:11,762 [main] INFO org.apache.pig.tools.pigstats.mapreduce.MRScriptState - Pig script settings are added to the job
2026-09-13 12:49:11,772 [main] INFO org.apache.hadoop.conf.Configuration.deprecation - mapred.job.reduce.markreset.buffer.percent is deprecated. Instead, use mapreduce.reduce.markreset.buffer.percent
2026-09-13 12:49:11,773 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.JobControlCompiler - mapred.job.reduce.markreset.buffer.percent is not set, set to default 0.3
2026-09-13 12:49:11,773 [main] INFO org.apache.hadoop.conf.Configuration.deprecation - mapred.output.compress is deprecated. Instead, use mapreduce.output.fileoutputformat.compress
2026-09-13 12:49:11,773 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.JobControlCompiler - Reduce phase detected, estimating # of required reducers.

2026-09-13 12:55:31,739 [main] INFO org.apache.hadoop.ipc.Client - Retrying connect to server: 0.0.0.0/0.0.0.0:10020. Already tried 5 time(s); retry policy is RetryUpToMaximumCountWithFixedSleep(maxRetries=10, sleepTime=1000 MILLISECONDS)
2026-09-13 12:55:34,794 [main] INFO org.apache.hadoop.ipc.Client - Retrying connect to server: 0.0.0.0/0.0.0.0:10020. Already tried 6 time(s); retry policy is RetryUpToMaximumCountWithFixedSleep(maxRetries=10, sleepTime=1000 MILLISECONDS)
2026-09-13 12:55:37,856 [main] INFO org.apache.hadoop.ipc.Client - Retrying connect to server: 0.0.0.0/0.0.0.0:10020. Already tried 7 time(s); retry policy is RetryUpToMaximumCountWithFixedSleep(maxRetries=10, sleepTime=1000 MILLISECONDS)
2026-09-13 12:55:40,906 [main] INFO org.apache.hadoop.ipc.Client - Retrying connect to server: 0.0.0.0/0.0.0.0:10020. Already tried 8 time(s); retry policy is RetryUpToMaximumCountWithFixedSleep(maxRetries=10, sleepTime=1000 MILLISECONDS)
2026-09-13 12:55:43,959 [main] INFO org.apache.hadoop.ipc.Client - Retrying connect to server: 0.0.0.0/0.0.0.0:10020. Already tried 9 time(s); retry policy is RetryUpToMaximumCountWithFixedSleep(maxRetries=10, sleepTime=1000 MILLISECONDS)
2026-09-13 12:55:46,117 [main] INFO org.apache.hadoop.mapred.ClientServiceDelegate - Application state is completed. FinalApplicationStatus=SUCCEEDED. Redirecting to job history server
2026-09-13 12:55:49,168 [main] INFO org.apache.hadoop.ipc.Client - Retrying connect to server: 0.0.0.0/0.0.0.0:10020. Already tried 0 time(s); retry policy is RetryUpToMaximumCountWithFixedSleep(maxRetries=10, sleepTime=1000 MILLISECONDS)
2026-09-13 12:55:52,208 [main] INFO org.apache.hadoop.ipc.Client - Retrying connect to server: 0.0.0.0/0.0.0.0:10020. Already tried 1 time(s); retry policy is RetryUpToMaximumCountWithFixedSleep(maxRetries=10, sleepTime=1000 MILLISECONDS)
2026-09-13 12:55:55,261 [main] INFO org.apache.hadoop.ipc.Client - Retrying connect to server: 0.0.0.0/0.0.0.0:10020. Already tried 2 time(s); retry policy is RetryUpToMaximumCountWithFixedSleep(maxRetries=10, sleepTime=1000 MILLISECONDS)
2026-09-13 12:55:58,311 [main] INFO org.apache.hadoop.ipc.Client - Retrying connect to server: 0.0.0.0/0.0.0.0:10020. Already tried 3 time(s); retry policy is RetryUpToMaximumCountWithFixedSleep(maxRetries=10, sleepTime=1000 MILLISECONDS)
2026-09-13 12:56:01,361 [main] INFO org.apache.hadoop.ipc.Client - Retrying connect to server: 0.0.0.0/0.0.0.0:10020. Already tried 4 time(s); retry policy is RetryUpToMaximumCountWithFixedSleep(maxRetries=10, sleepTime=1000 MILLISECONDS)
2026-09-13 12:56:04,411 [main] INFO org.apache.hadoop.ipc.Client - Retrying connect to server: 0.0.0.0/0.0.0.0:10020. Already tried 5 time(s); retry policy is RetryUpToMaximumCountWithFixedSleep(maxRetries=10, sleepTime=1000 MILLISECONDS)
2026-09-13 12:56:07,472 [main] INFO org.apache.hadoop.ipc.Client - Retrying connect to server: 0.0.0.0/0.0.0.0:10020. Already tried 6 time(s); retry policy is RetryUpToMaximumCountWithFixedSleep(maxRetries=10, sleepTime=1000 MILLISECONDS)
2026-09-13 12:56:10,519 [main] INFO org.apache.hadoop.ipc.Client - Retrying connect to server: 0.0.0.0/0.0.0.0:10020. Already tried 7 time(s); retry policy is RetryUpToMaximumCountWithFixedSleep(maxRetries=10, sleepTime=1000 MILLISECONDS)
2026-09-13 12:56:13,571 [main] INFO org.apache.hadoop.ipc.Client - Retrying connect to server: 0.0.0.0/0.0.0.0:10020. Already tried 8 time(s); retry policy is RetryUpToMaximumCountWithFixedSleep(maxRetries=10, sleepTime=1000 MILLISECONDS)
2026-09-13 12:56:16,630 [main] INFO org.apache.hadoop.ipc.Client - Retrying connect to server: 0.0.0.0/0.0.0.0:10020. Already tried 9 time(s); retry policy is RetryUpToMaximumCountWithFixedSleep(maxRetries=10, sleepTime=1000 MILLISECONDS)
2026-09-13 12:56:18,768 [main] INFO org.apache.hadoop.mapred.ClientServiceDelegate - Application state is completed. FinalApplicationStatus=SUCCEEDED. Redirecting to job history server
2026-09-13 12:56:21,811 [main] INFO org.apache.hadoop.ipc.Client - Retrying connect to server: 0.0.0.0/0.0.0.0:10020. Already tried 0 time(s); retry policy is RetryUpToMaximumCountWithFixedSleep(maxRetries=10, sleepTime=1000 MILLISECONDS)
2026-09-13 12:56:24,859 [main] INFO org.apache.hadoop.ipc.Client - Retrying connect to server: 0.0.0.0/0.0.0.0:10020. Already tried 1 time(s); retry policy is RetryUpToMaximumCountWithFixedSleep(maxRetries=10, sleepTime=1000 MILLISECONDS)
Terminate batch job (Y/N)? N
```

Output

```
C:\BDLFinalProject\PigAnalysis\scripts>hadoop fs -cat Top10MostReviewedByPublisher.txt/part-r-00000 | more
3|3|0967389395|Restoring Intimacy: The Patient's Guide to Maintaining Relationships During Depression|1
AM|AM|8448465114|La tahona de Froil|in|1
BQ|BQ|2894060009|Ang|line de Montbrun|2
LP|LP|2253010073|Therese Raquin|4
MI|MI|347835630X|Das Dilbert- Prinzip.|2
MI|MI|3478245605|Reich durch Beziehungen. Durch die richtigen Kontakte zum Erfolg.|1
P3|P3|0752585398|Practical Rice Dishes|2
QC|QC|8477271666|Estiu tot l'any (M|nima de butxaca)|1
SM|SM|8434852861|Campos de fresas (Gran angular. Alerta roja)|2
Sm|Sm|8434808633|Fray Perico Y Su Borrigo/Brother Perico and His Donkey|1
US|US|0965712516|The Missionary's Catechism|1
AAA|AAA|1562514423|AAA 2001 Britain TravelBook (Aaa Britain Travelbook)|2
AAA|AAA|1562512617|AAA PET BOOK|1
AAA|AAA|1562516809|AAA Europe TravelBook : The Guide to Premier Destinations 2002 Edition|1
AAA|AAA|1562511785|AAA 1996 NORTH AMERICAN ROAD ATLAS (Aaa North American Road Atlas)|1
ACT|ACT|0937734608|College Planning Search Book|1
AGS|AGS|0806713560|Parenting young children: Helpful strategies based on Systematic Training for Effective Parenting (STEP) for parents of children under si
x|2
ASA|ASA|9724111229|Os comedores de p|rolas (Finisterra)|1
ASA|ASA|97241117588|Os passaros de seda (Finisterra)|1
Ace|Ace|0441010288|The Lost Years of Merlin|3
Ace|Ace|0441000231|Isaac Asimov's Inferno|3
Ace|Ace|0441011322|Conqueror's Moon (The Boreal Moon Tale Book 1)|1
Ace|Ace|0441011985|For Those Who Fell|1
Ace|Ace|0441011829|Myth Alliances (Myth Adventures Series)|1
Ace|Ace|0441003303|Archangel|1
Ace|Ace|0441011993|Furies Of Calderon|1
Ace|Ace|0441012159|The Nimble Man (Menagerie)|1
Ace|Ace|0441116868|Conjure Wife|1
Ace|Ace|0441011799|Singularity Sky|1
Adp|Adp|0885660838|Bonheur d Occasion (Qu|bec 10/10 [i.e. dix sur dix])|4
Alk|Alk|9060139097|Onze Nederlandse stoomlocomotieven in woord en beeld|1
BBC|BBC|0563534753|The Life Laundry: How to De-Junk Your Life|7
BBC|BBC|0563534222|Terry Wogan: Is It Me|6
BBC|BBC|0563551011|Changing Rooms|5
BBC|BBC|056353821X|Steve Redgrave: A Golden Age|4
BBC|BBC|0563362499|Delia Smith's Complete Cookery Course|3
```

4. Calculate the Top 10 Highest Rated Books: Returns the Top 10 highest-rated books based on their average rating.

Script

```
-- Top 10 individual ratings in the merged dataset
books = LOAD 'hdfs:///BDLFinalProject/data/bookDataset.csv' USING PigStorage(',') AS
(isbn:chararray, title:chararray, author:chararray, year:int, publisher:chararray,
userid:chararray, rating:float);
filtered = FILTER books BY isbn IS NOT NULL AND isbn != 'ISBN' AND rating IS NOT NULL;
sorted = ORDER filtered BY rating DESC;
top10 = LIMIT sorted 10;
STORE top10 INTO 'Top10Rated.txt' using PigStorage('|');
```

Execution

```
C:\BDLFinalProject\PigAnalysis\scripts>pig -x mapreduce top10Rated.pig
2026-09-13 13:21:04,367 INFO pig.ExecTypeProvider: Trying ExecType : LOCAL
2026-09-13 13:21:04,367 INFO pig.ExecTypeProvider: Trying ExecType : MAPREDUCE
2026-09-13 13:21:04,367 INFO pig.ExecTypeProvider: Picked MAPREDUCE as the ExecType
2026-09-13 13:21:04,588 [main] INFO org.apache.pig.Main - Apache Pig version 0.18.0 (r1926484) compiled Jun 16 2025, 10:38:21
2026-09-13 13:21:04,588 [main] INFO org.apache.pig.Main - Logging error messages to: C:\hadoop\logs\pig_1789284064586.log
2026-09-13 13:21:04,945 [main] INFO org.apache.pig.impl.util.Utils - Default bootstrap file C:\Users\absar/.pigbootstrap not found
2026-09-13 13:21:04,988 [main] INFO org.apache.hadoop.conf.Configuration.deprecation - mapred.job.tracker is deprecated. Instead, use mapreduce.jobtracker.address
2026-09-13 13:21:04,988 [main] INFO org.apache.pig.backend.hadoop.executionengine.HExecutionEngine - Connecting to hadoop file system at: hdfs://localhost:9000
2026-09-13 13:21:05,345 [main] INFO org.apache.pig.PigServer - Pig Script ID for the session: PIG-top10Rated.pig-e7cba65e-f072-4b18-81c1-6bc11954d9f7
2026-09-13 13:21:05,345 [main] WARN org.apache.pig.PigServer - ATS is disabled since yarn.timeline-service.enabled set to false
2026-09-13 13:21:05,842 [main] INFO org.apache.hadoop.conf.Configuration.deprecation - mapred.textoutputformat.separator is deprecated. Instead, use mapreduce.textoutputformat.separator
2026-09-13 13:21:05,863 [main] INFO org.apache.pig.tools.pigstats.ScriptState - Pig features used in the script: ORDER_BY,FILTER,LIMIT
2026-09-13 13:21:05,872 [main] INFO org.apache.pig.data.SchemaTupleBackend - Key [pig.schematuple] was not set... will not generate code.
2026-09-13 13:21:05,918 [main] INFO org.apache.pig.newplan.logical.optimizer.LogicalPlanOptimizer - {RULES_ENABLED=[AddForEach, ColumnMapKeyPrune, FilterConstantCalculator, ForEachConstantCalculator, GroupByConstParallelSetter, LimitOptimizer, LoadTypeCastInsertter, MergeFilter, MergeForEach, NestedLimitOptimizer, PartitionFilterOptimizer, PredicatePushdownOptimizer, PushDownForEachFlatten, PushUpFilter, SplitConstantCalculator, SplitFilter, StreamTypeCastInsertter]}
2026-09-13 13:21:05,957 [main] INFO org.apache.pig.impl.util.SpillableMemoryManager - Selected heap (PS Old Gen) of size 699400192 to monitor. collectionUsageThreshold = 489580128, usageThreshold = 489580128
2026-09-13 13:21:06,006 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MRCompiler - File concatenation threshold: 100 optimistic? false
2026-09-13 13:21:06,092 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.SecondaryKeyOptimizerMR - Using Secondary Key Optimization for MapReduce node scope=41
2026-09-13 13:21:06,107 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MultiQueryOptimizer - MR plan size before optimization: 4
2026-09-13 13:21:06,107 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MultiQueryOptimizer - MR plan size after optimization: 4
2026-09-13 13:21:06,188 [main] INFO org.apache.hadoop.yarn.client.RMProxy - Connecting to ResourceManager at /0.0.0.0:8032
2026-09-13 13:21:06,367 [main] INFO org.apache.hadoop.conf.Configuration.deprecation - yarn.resourcemanager.system-metrics-publisher.enabled is deprecated. Instead, use yarn.system-metrics-publisher.enabled
2026-09-13 13:21:06,382 [main] INFO org.apache.pig.tools.pigstats.mapreduce.MRScriptState - Pig script settings are added to the job
2026-09-13 13:21:06,388 [main] INFO org.apache.hadoop.conf.Configuration.deprecation - mapred.job.reduce.markreset.buffer.percent is deprecated. Instead, use mapreduce.reduce.markreset.buffer.percent
2026-09-13 13:21:06,388 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.JobControlCompiler - mapred.job.reduce.markreset.buffer.percent is not set, set to default 0.3
2026-09-13 13:21:06,392 [main] INFO org.apache.hadoop.conf.Configuration.deprecation - mapred.output.compress is deprecated. Instead, use mapreduce.output.fileoutputformat.compress
2026-09-13 13:21:06,394 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.JobControlCompiler - This job cannot be converted run in-p

2026-09-13 13:38:02,592 [main] INFO org.apache.hadoop.ipc.Client - Retrying connect to server: 0.0.0.0/0.0.0.0:10020. Already tried 0 time(s); retry policy is RetryUpToMaximumCountWithFixedSleep(maxRetries=10, sleepTime=1000 MILLISECONDS)
2026-09-13 13:38:05,639 [main] INFO org.apache.hadoop.ipc.Client - Retrying connect to server: 0.0.0.0/0.0.0.0:10020. Already tried 1 time(s); retry policy is RetryUpToMaximumCountWithFixedSleep(maxRetries=10, sleepTime=1000 MILLISECONDS)
2026-09-13 13:38:08,695 [main] INFO org.apache.hadoop.ipc.Client - Retrying connect to server: 0.0.0.0/0.0.0.0:10020. Already tried 2 time(s); retry policy is RetryUpToMaximumCountWithFixedSleep(maxRetries=10, sleepTime=1000 MILLISECONDS)
2026-09-13 13:38:11,747 [main] INFO org.apache.hadoop.ipc.Client - Retrying connect to server: 0.0.0.0/0.0.0.0:10020. Already tried 3 time(s); retry policy is RetryUpToMaximumCountWithFixedSleep(maxRetries=10, sleepTime=1000 MILLISECONDS)
2026-09-13 13:38:14,800 [main] INFO org.apache.hadoop.ipc.Client - Retrying connect to server: 0.0.0.0/0.0.0.0:10020. Already tried 4 time(s); retry policy is RetryUpToMaximumCountWithFixedSleep(maxRetries=10, sleepTime=1000 MILLISECONDS)
2026-09-13 13:38:17,854 [main] INFO org.apache.hadoop.ipc.Client - Retrying connect to server: 0.0.0.0/0.0.0.0:10020. Already tried 5 time(s); retry policy is RetryUpToMaximumCountWithFixedSleep(maxRetries=10, sleepTime=1000 MILLISECONDS)
2026-09-13 13:38:20,910 [main] INFO org.apache.hadoop.ipc.Client - Retrying connect to server: 0.0.0.0/0.0.0.0:10020. Already tried 6 time(s); retry policy is RetryUpToMaximumCountWithFixedSleep(maxRetries=10, sleepTime=1000 MILLISECONDS)
2026-09-13 13:38:23,960 [main] INFO org.apache.hadoop.ipc.Client - Retrying connect to server: 0.0.0.0/0.0.0.0:10020. Already tried 7 time(s); retry policy is RetryUpToMaximumCountWithFixedSleep(maxRetries=10, sleepTime=1000 MILLISECONDS)
2026-09-13 13:38:27,014 [main] INFO org.apache.hadoop.ipc.Client - Retrying connect to server: 0.0.0.0/0.0.0.0:10020. Already tried 8 time(s); retry policy is RetryUpToMaximumCountWithFixedSleep(maxRetries=10, sleepTime=1000 MILLISECONDS)
2026-09-13 13:38:30,064 [main] INFO org.apache.hadoop.ipc.Client - Retrying connect to server: 0.0.0.0/0.0.0.0:10020. Already tried 9 time(s); retry policy is RetryUpToMaximumCountWithFixedSleep(maxRetries=10, sleepTime=1000 MILLISECONDS)
2026-09-13 13:38:32,223 [main] INFO org.apache.hadoop.mapred.ClientServiceDelegate - Application state is completed. FinalApplicationStatus=SUCCEEDED. Redirecting to job history server
2026-09-13 13:38:35,278 [main] INFO org.apache.hadoop.ipc.Client - Retrying connect to server: 0.0.0.0/0.0.0.0:10020. Already tried 0 time(s); retry policy is RetryUpToMaximumCountWithFixedSleep(maxRetries=10, sleepTime=1000 MILLISECONDS)
2026-09-13 13:38:38,331 [main] INFO org.apache.hadoop.ipc.Client - Retrying connect to server: 0.0.0.0/0.0.0.0:10020. Already tried 1 time(s); retry policy is RetryUpToMaximumCountWithFixedSleep(maxRetries=10, sleepTime=1000 MILLISECONDS)
2026-09-13 13:38:41,383 [main] INFO org.apache.hadoop.ipc.Client - Retrying connect to server: 0.0.0.0/0.0.0.0:10020. Already tried 2 time(s); retry policy is RetryUpToMaximumCountWithFixedSleep(maxRetries=10, sleepTime=1000 MILLISECONDS)
2026-09-13 13:38:44,444 [main] INFO org.apache.hadoop.ipc.Client - Retrying connect to server: 0.0.0.0/0.0.0.0:10020. Already tried 3 time(s); retry policy is RetryUpToMaximumCountWithFixedSleep(maxRetries=10, sleepTime=1000 MILLISECONDS)
2026-09-13 13:38:47,496 [main] INFO org.apache.hadoop.ipc.Client - Retrying connect to server: 0.0.0.0/0.0.0.0:10020. Already tried 4 time(s); retry policy is RetryUpToMaximumCountWithFixedSleep(maxRetries=10, sleepTime=1000 MILLISECONDS)
2026-09-13 13:38:50,555 [main] INFO org.apache.hadoop.ipc.Client - Retrying connect to server: 0.0.0.0/0.0.0.0:10020. Already tried 5 time(s); retry policy is RetryUpToMaximumCountWithFixedSleep(maxRetries=10, sleepTime=1000 MILLISECONDS)
2026-09-13 13:38:53,611 [main] INFO org.apache.hadoop.ipc.Client - Retrying connect to server: 0.0.0.0/0.0.0.0:10020. Already tried 6 time(s); retry policy is RetryUpToMaximumCountWithFixedSleep(maxRetries=10, sleepTime=1000 MILLISECONDS)
2026-09-13 13:38:56,663 [main] INFO org.apache.hadoop.ipc.Client - Retrying connect to server: 0.0.0.0/0.0.0.0:10020. Already tried 7 time(s); retry policy is RetryUpToMaximumCountWithFixedSleep(maxRetries=10, sleepTime=1000 MILLISECONDS)
Terminate batch job (Y/N)? N
```

Output

```
C:\BDLFinalProject\PigAnalysis\scripts>hadoop fs -cat Top10Rated.txt/part-r-00000
0878510370|Good Housekeeping Illustrated Cookbook|Coulson|1980|Harpercollins|39281|10.0
0898624401|Friedman's Fables (with Booklet)|Edwin H. Friedman|1990|The Guilford Press|39281|10.0
0964427907|The Inclusive New Testament|Priests for Equality|1994|Priests for Equality|39281|10.0
0060934417|Bel Canto: A Novel|Ann Patchett|2002|Perennial|154064|10.0
1401300057|Firehouse|David Halberstam|2002|Hyperion Press|39281|10.0
1558531025|Life's Little Instruction Book (Life's Little Instruction Books (Paperback))|H. Jackson Brown|1991|Thomas Nelson|39281|10.0
1565840003|Race: How Blacks and Whites Think and Feel about the American Obsession|Studs Terkel|1992|W W Norton & Co Inc|39281|10.0
157866005X|Cats' Letters to Santa|Bill Adler|2004|Galahad|39281|10.0
0441023827|The Anubis Gates|Tim Powers|1984|Ace Books|39287|10.0
1582970734|2002 Novel & Short Story Writers Market (Novel and Short Story Writer's Market)|Anne Bowling|2001|Writer's Digest Books|39287|10.0
```

5. Calculate the Top 10 Highest Rated Books by Publisher: Returns the Top 10 highest-rated books for each publisher based on their average rating.

Script

```
-- Top 10 highest-rated records for each publisher
books = LOAD 'hdfs:///BDLFinalProject/data/bookDataset.csv' USING PigStorage(',') AS
(isbn:chararray, title:chararray, author:chararray, year:int, publisher:chararray,
userid:chararray, rating:float);
filtered = FILTER books BY isbn IS NOT NULL AND isbn != 'ISBN' AND publisher IS NOT NULL
AND publisher != '' AND rating IS NOT NULL;
grp = GROUP filtered BY publisher;
top_each = FOREACH grp { ordered = ORDER filtered BY rating DESC; limited = LIMIT ordered
10; GENERATE group AS publisher, FLATTEN(limited); };
STORE top_each INTO 'Top10RatedByPublisher.txt' using PigStorage('|');
```

Execution

```
C:\BDLFinalProject\PigAnalysis\scripts>pig -x mapreduce top10RatedByPublisher.pig
2026-09-13 11:56:57,359 INFO pig.ExecTypeProvider: Trying ExecType : LOCAL
2026-09-13 11:56:57,363 INFO pig.ExecTypeProvider: Trying ExecType : MAPREDUCE
2026-09-13 11:56:57,363 INFO pig.ExecTypeProvider: Picked MAPREDUCE as the ExecType
2026-09-13 11:56:57,552 [main] INFO org.apache.pig.Main - Apache Pig version 0.18.0 (r1926484) compiled Jun 16 2025, 10:38:21
2026-09-13 11:56:57,552 [main] INFO org.apache.pig.Main - Logging error messages to: C:\hadoop\logs\pig_1789279017550.log
2026-09-13 11:56:57,868 [main] INFO org.apache.pig.impl.util.Util - Default bootstrap file C:\Users\absar/.pigbootstrap not found
2026-09-13 11:56:57,923 [main] INFO org.apache.hadoop.conf.Configuration.deprecation - mapred.job.tracker is deprecated. Instead, use mapreduce.jobtracker
address
2026-09-13 11:56:57,924 [main] INFO org.apache.pig.backend.hadoop.executionengine.HExecutionEngine - Connecting to hadoop file system at: hdfs://localhost
9000
2026-09-13 11:56:58,330 [main] INFO org.apache.pig.PigServer - Pig Script ID for the session: PIG-top10RatedByPublisher.pig-c57bdf5-4e0e-48a6-b7b8-f2d3e8
0dfb7
2026-09-13 11:56:58,330 [main] WARN org.apache.pig.PigServer - ATS is disabled since yarn.timeline-service.enabled set to false
2026-09-13 11:56:58,850 [main] INFO org.apache.hadoop.conf.Configuration.deprecation - mapred.textoutputformat.separator is deprecated. Instead, use mapre
uce.output.textoutputformat.separator
2026-09-13 11:56:58,872 [main] INFO org.apache.pig.tools.pigstats.ScriptState - Pig features used in the script: GROUP_BY,FILTER
2026-09-13 11:56:58,891 [main] INFO org.apache.pig.data.SchemaTupleBackend - Key [pig.schematuple] was not set... will not generate code.
2026-09-13 11:56:58,987 [main] INFO org.apache.pig.newplan.logical.optimizer.LogicalPlanOptimizer - {RULES_ENABLED=[AddForEach, ColumnMapKeyPrune, FilterC
onstantCalculator, ForEachConstantCalculator, GroupByConstParallelSetter, LimitOptimizer, LoadTypeCastInserter, MergeFilter, MergeForEach, NestedLimitOptimi
zer, PartitionFilterOptimizer, PredicatePushdownOptimizer, PushDownForEachFlatten, PushUpFilter, SplitConstantCalculator, SplitFilter, StreamTypeCastInserter
]}
2026-09-13 11:56:58,955 [main] INFO org.apache.pig.impl.util.SpillableMemoryManager - Selected heap (PS Old Gen) of size 699400192 to monitor. collectionU
ageThreshold = 489580128, usageThreshold = 489580128
2026-09-13 11:56:58,995 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MRCompiler - File concatenation threshold: 100 optimistic
false
2026-09-13 11:56:59,018 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MultiQueryOptimizer - MR plan size before optimization: 1
2026-09-13 11:56:59,018 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MultiQueryOptimizer - MR plan size after optimization: 1
2026-09-13 11:56:59,068 [main] INFO org.apache.hadoop.yarn.client.RMPProxy - Connecting to ResourceManager at /0.0.0.0:8032
2026-09-13 11:56:59,235 [main] INFO org.apache.hadoop.conf.Configuration.deprecation - yarn.resourcemanager.system-metrics-publisher.enabled is deprecated
Instead, use yarn.system-metrics-publisher.enabled
2026-09-13 11:56:59,243 [main] INFO org.apache.pig.tools.pigstats.mapreduce.MRScriptState - Pig script settings are added to the job
2026-09-13 11:56:59,260 [main] INFO org.apache.hadoop.conf.Configuration.deprecation - mapred.job.reduce.markreset.buffer.percent is deprecated. Instead,
use mapreduce.reduce.markreset.buffer.percent
2026-09-13 11:56:59,260 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.JobControlCompiler - mapred.job.reduce.markreset.buffer.p
ercent is not set, set to default 0.3
2026-09-13 11:56:59,262 [main] INFO org.apache.hadoop.conf.Configuration.deprecation - mapred.output.compress is deprecated. Instead, use mapreduce.output
fileoutputformat.compress
2026-09-13 11:56:59,266 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.JobControlCompiler - Reduce phase detected, estimating #
of required reducers.
```

```

2026-09-13 12:04:19,013 [main] INFO org.apache.hadoop.ipc.Client - Retrying connect to server: 0.0.0.0/0.0.0.0:10020. Already tried 1 time(s); retry policy
is RetryUpToMaximumCountWithFixedSleep(maxRetries=10, sleepTime=1000 MILLISECONDS)
2026-09-13 12:04:22,063 [main] INFO org.apache.hadoop.ipc.Client - Retrying connect to server: 0.0.0.0/0.0.0.0:10020. Already tried 2 time(s); retry policy
is RetryUpToMaximumCountWithFixedSleep(maxRetries=10, sleepTime=1000 MILLISECONDS)
2026-09-13 12:04:25,117 [main] INFO org.apache.hadoop.ipc.Client - Retrying connect to server: 0.0.0.0/0.0.0.0:10020. Already tried 3 time(s); retry policy
is RetryUpToMaximumCountWithFixedSleep(maxRetries=10, sleepTime=1000 MILLISECONDS)
2026-09-13 12:04:28,164 [main] INFO org.apache.hadoop.ipc.Client - Retrying connect to server: 0.0.0.0/0.0.0.0:10020. Already tried 4 time(s); retry policy
is RetryUpToMaximumCountWithFixedSleep(maxRetries=10, sleepTime=1000 MILLISECONDS)
2026-09-13 12:04:31,213 [main] INFO org.apache.hadoop.ipc.Client - Retrying connect to server: 0.0.0.0/0.0.0.0:10020. Already tried 5 time(s); retry policy
is RetryUpToMaximumCountWithFixedSleep(maxRetries=10, sleepTime=1000 MILLISECONDS)
2026-09-13 12:04:34,259 [main] INFO org.apache.hadoop.ipc.Client - Retrying connect to server: 0.0.0.0/0.0.0.0:10020. Already tried 6 time(s); retry policy
is RetryUpToMaximumCountWithFixedSleep(maxRetries=10, sleepTime=1000 MILLISECONDS)
2026-09-13 12:04:37,313 [main] INFO org.apache.hadoop.ipc.Client - Retrying connect to server: 0.0.0.0/0.0.0.0:10020. Already tried 7 time(s); retry policy
is RetryUpToMaximumCountWithFixedSleep(maxRetries=10, sleepTime=1000 MILLISECONDS)
2026-09-13 12:04:40,363 [main] INFO org.apache.hadoop.ipc.Client - Retrying connect to server: 0.0.0.0/0.0.0.0:10020. Already tried 8 time(s); retry policy
is RetryUpToMaximumCountWithFixedSleep(maxRetries=10, sleepTime=1000 MILLISECONDS)
2026-09-13 12:04:43,408 [main] INFO org.apache.hadoop.ipc.Client - Retrying connect to server: 0.0.0.0/0.0.0.0:10020. Already tried 9 time(s); retry policy
is RetryUpToMaximumCountWithFixedSleep(maxRetries=10, sleepTime=1000 MILLISECONDS)
2026-09-13 12:04:45,561 [main] INFO org.apache.hadoop.mapred.ClientServiceDelegate - Application state is completed. FinalApplicationStatus=SUCCEEDED. Redi
recting to job history server
2026-09-13 12:04:48,608 [main] INFO org.apache.hadoop.ipc.Client - Retrying connect to server: 0.0.0.0/0.0.0.0:10020. Already tried 0 time(s); retry policy
is RetryUpToMaximumCountWithFixedSleep(maxRetries=10, sleepTime=1000 MILLISECONDS)
2026-09-13 12:04:51,656 [main] INFO org.apache.hadoop.ipc.Client - Retrying connect to server: 0.0.0.0/0.0.0.0:10020. Already tried 1 time(s); retry policy
is RetryUpToMaximumCountWithFixedSleep(maxRetries=10, sleepTime=1000 MILLISECONDS)
2026-09-13 12:04:54,703 [main] INFO org.apache.hadoop.ipc.Client - Retrying connect to server: 0.0.0.0/0.0.0.0:10020. Already tried 2 time(s); retry policy
is RetryUpToMaximumCountWithFixedSleep(maxRetries=10, sleepTime=1000 MILLISECONDS)
2026-09-13 12:04:57,752 [main] INFO org.apache.hadoop.ipc.Client - Retrying connect to server: 0.0.0.0/0.0.0.0:10020. Already tried 3 time(s); retry policy
is RetryUpToMaximumCountWithFixedSleep(maxRetries=10, sleepTime=1000 MILLISECONDS)
2026-09-13 12:05:00,802 [main] INFO org.apache.hadoop.ipc.Client - Retrying connect to server: 0.0.0.0/0.0.0.0:10020. Already tried 4 time(s); retry policy
is RetryUpToMaximumCountWithFixedSleep(maxRetries=10, sleepTime=1000 MILLISECONDS)
2026-09-13 12:05:03,857 [main] INFO org.apache.hadoop.ipc.Client - Retrying connect to server: 0.0.0.0/0.0.0.0:10020. Already tried 5 time(s); retry policy
is RetryUpToMaximumCountWithFixedSleep(maxRetries=10, sleepTime=1000 MILLISECONDS)
2026-09-13 12:05:06,909 [main] INFO org.apache.hadoop.ipc.Client - Retrying connect to server: 0.0.0.0/0.0.0.0:10020. Already tried 6 time(s); retry policy
is RetryUpToMaximumCountWithFixedSleep(maxRetries=10, sleepTime=1000 MILLISECONDS)
2026-09-13 12:05:09,961 [main] INFO org.apache.hadoop.ipc.Client - Retrying connect to server: 0.0.0.0/0.0.0.0:10020. Already tried 7 time(s); retry policy
is RetryUpToMaximumCountWithFixedSleep(maxRetries=10, sleepTime=1000 MILLISECONDS)
2026-09-13 12:05:13,008 [main] INFO org.apache.hadoop.ipc.Client - Retrying connect to server: 0.0.0.0/0.0.0.0:10020. Already tried 8 time(s); retry policy
is RetryUpToMaximumCountWithFixedSleep(maxRetries=10, sleepTime=1000 MILLISECONDS)
Terminate batch job (Y/N)? N

```

Output

```

C:\BDLFinalProject\PigAnalysis\scripts>hdfs dfs -cat Top10RatedByPublisher.txt/part-r-00000 | more
3|0967389305|Restoring Intimacy: The Patient's Guide to Maintaining Relationships During Depression|Drew Pinsky|1999|3|265276|8.0
AM|8440465114|La tahona de Froilán|in|Jos|António Pet Possel|1990|AM|49144|5.0
BQ|2894060009|Ang|Line de Montbrun|Laure Conan|1988|BQ|32793|0.0
BQ|2894060009|Ang|Line de Montbrun|Laure Conan|1988|BQ|146465|0.0
LP|2253010073|Therese Raquin|Emile Zola|LP|79933|10.0
LP|2253010073|Therese Raquin|Emile Zola|LP|222912|10.0
LP|2253010073|Therese Raquin|Emile Zola|LP|215947|10.0
LP|2253010073|Therese Raquin|Emile Zola|LP|23902|8.0
MI|347835630X|Das Dilbert- Prinzip.|Scott Adams|1997|MI|107951|9.0
MI|347835630X|Das Dilbert- Prinzip.|Scott Adams|1997|MI|95757|8.0
MI|3478245605|Reich durch Beziehungen. Durch die richtigen Kontakte zum Erfolg.|Alfred J. Kremer|2000|MI|63650|3.0
P3|0752585398|Practical Rice Dishes|P3|2002|P3|204766|3.0
P3|0752585398|Practical Rice Dishes|P3|2002|P3|148744|0.0
QC|8477271666|Estiu tot l'any (Màxim de butxaca)|Anton M Espadaler|1996|QC|23419|8.0
SM|8434852861|Campos de fresas (Gran angular. Alerta roja)|Jordi Sierra i Fabra|1997|SM|34827|10.0
SM|8434852861|Campos de fresas (Gran angular. Alerta roja)|Jordi Sierra i Fabra|1997|SM|137522|6.0
Sm|8434808633|Fray Perico Y Su Borrico|Brother Perico and His Donkey|Juan Munoz Martin|1984|Sm|136579|0.0
US|0965712516|The Missionary's Catechism|Russell L. Ford|1998|US|1297|0.0
AAA|1562514423|AAA 2001 Britain TravelBook (Aaa Britain Travelbook)|AAA Publishing|2001|AAA|255489|7.0
AAA|1562512617|AAA PET BOOK|AAA|1997|AAA|84024|7.0
AAA|1562516809|AAA Europe TravelBook : The Guide to Premier Destinations 2002 Edition|AAA Publishing|2002|AAA|59189|0.0
AAA|1562511785|AAA 1996 NORTH AMERICAN ROAD ATLAS (Aaa North American Road Atlas)|American Automobile Association|1995|AAA|230522|0.0
AAA|1562514423|AAA 2001 Britain TravelBook (Aaa Britain Travelbook)|AAA Publishing|2001|AAA|209156|0.0
ACT|8937734608|College Planning Search Book|American College Testing Program|1987|ACT|185233|0.0
AGS|0886713560|Parenting young children: Helpful strategies based on Systematic Training for Effective Parenting (STEP) for parents of children under six|De
n C Dinkmeyer|1989|AGS|153662|8.0
AGS|0886713560|Parenting young children: Helpful strategies based on Systematic Training for Effective Parenting (STEP) for parents of children under six|De
n C Dinkmeyer|1989|AGS|153831|0.0
ASA|9724117588|Os passavos de seda (Finisterra)|Rosa Lobato de Faria|1996|ASA|71612|8.0
ASA|9724111229|Os comedores de pã-rolas (Finisterra)|João Aguiar|1992|ASA|271705|8.0
Ace|0441011322|Conqueror's Moon (The Boreal Moon Tale Book 1)|Julian May|2004|Ace|98391|10.0
Ace|0441012159|The Nimble Man (Menagerie)|Christopher Golden|2004|Ace|98391|10.0
Ace|0441011993|Furies Of Calderon|Jim Butcher|2004|Ace|98391|10.0
Ace|0441011985|For Those Who Fell|William C. Dietz|2004|Ace|98391|10.0
Ace|0441000231|Isaac Asimov's Inferno|Roger MacBride Allen|1994|Ace|33179|9.0
Ace|0441116868|Conjure Wife|Fritz Leiber|1977|Ace|2276|9.0
Ace|0441000231|Isaac Asimov's Inferno|Roger MacBride Allen|1994|Ace|64469|8.0
Ace|0441010288|The Lost Years of Merlin|T. A. Barron|2002|Ace|167722|7.0

```

Description:

In this lab report, I performed MapReduce and Pig data analysis on the Goodreads Book Recommendation dataset. First, I merged the required CSV files using ISBN and uploaded the final dataset to HDFS. Then I applied MapReduce and Pig jobs for data analysis. The outputs of all the tasks are presented in this report with screenshots.